

CHUKKA VENUGOPALAM

Visakhapatnam, India | 9908214586 | chukkavenugopalam@gmail.com | [Portfolio](#) | [GitHub](#) | [LeetCode](#)

Full-stack and AI/ML-focused Computer Science undergraduate (CGPA 9.45/10) who builds products end-to-end — from concept graphs and AI-assisted diagnostics to civic-tech and polling platforms. Motivated by turning ideas people actually use into working systems, with strong foundations in DSA, OS, DBMS, and system design.

EDUCATION

Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam

2024 – 2028

B.Tech in Computer Science Engineering

CGPA: 9.45 / 10

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, HTML, CSS

Frameworks & Backend: Next.js, React, FastAPI, REST APIs

Database & Cloud: PostgreSQL, Supabase, Vercel, Render

AI/ML: Gemini API, Prompt Engineering, LLM Integration

Core CS & Tools: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, System Design, Git, GitHub, Figma

FLAGSHIP PROJECTS

Concept Intelligence Platform (*Landing Page Only — Product Not Yet Functional*) | Next.js, FastAPI, PostgreSQL (Supabase), Gemini 1.5 Flash, Python

[Live](#) | [GitHub](#)

- Architected an AI-driven diagnostic platform to measure conceptual understanding of DSA topics, going beyond binary correct/incorrect grading used by platforms like LeetCode
- Designed a hybrid system where an LLM handles natural-language concept extraction while a deterministic Python engine performs gap detection, weighted scoring, and topological sorting — avoiding black-box grading
- Manually curated an ~80-node expert concept graph across 4 MVP topics (Arrays & Hashing, Linked Lists, Trees, Graphs) with explicit knowledge-dependency relationships
- Completed full system design, database schema, API design, and a 6-week implementation roadmap

Silicon Valley Learning OS (*Active Development*) | Next.js, TypeScript, Cloud Deployment

[Live](#) | [GitHub](#)

- Architecting an AI-powered civic services platform for digital governance and public service accessibility
- Building responsive, authenticated full-stack architecture; live preview deployed on Vercel

PulseVote (*Landing Page Only — Product Not Yet Functional*) | Next.js, Cloud Deployment

[Live](#) | [GitHub](#)

- Designed the product concept and landing page for an AI-assisted polling platform focused on opinion collection, trend analysis, and data visualization
- Planned full-stack architecture (Next.js, authentication, cloud deployment); core functionality in development

EXPERIENCE

Software Developer Intern — DriftSync Solutions Pvt. Ltd. (*May – Jun 2026*)

- Contributed UI/UX improvements and bug fixes for EatssApp, a rural food delivery platform, and assisted with company website development
- Collaborated within a 3-member engineering team on Next.js and Android app development workflows under senior developer guidance

OTHER PROJECTS

OS Scheduling Simulator & Analyzer | HTML, CSS, JavaScript

[Live](#) | [GitHub](#)

- Developed a simulator for CPU scheduling algorithms (FCFS, SJF, SRTF, Priority, Round Robin) supporting preemptive and non-preemptive modes; built Gantt chart visualizations and a comparison engine for Waiting/Turnaround/Completion time

Interactive Graph Algorithm Visualizer | HTML5 Canvas, JavaScript

[Live](#) | [GitHub](#)

- Built an interactive visualizer for graph traversal, shortest-path, and MST algorithms (BFS, DFS, Dijkstra, Floyd-Warshall, Prim's, Kruskal's) with real-time animations of internal data structures

Page Replacement Algorithm Visualizer | HTML, CSS, JavaScript

[Live](#) | [GitHub](#)

- Built a step-by-step simulator comparing FIFO, LRU, and Optimal page replacement algorithms with live page-fault and hit-ratio tracking

ACHIEVEMENTS

- 1st Prize, Project Warzone — college-level project expo
- Qualified for Smart India Hackathon (SIH) submission via Innovex internal selection event
- Participant, SnowHack 2025
- Solved 160+ problems on LeetCode (C++) across Arrays, Strings, DP, Backtracking, Hashing, and more

INTERESTS

Reading novels, manhwa & manga; books on stock markets and finance; studying business strategy